

MATRIX FACTORIZATIONS

$$1. \quad A = LU = \begin{pmatrix} \text{lower triangular } L \\ \text{1's on the diagonal} \end{pmatrix} \begin{pmatrix} \text{upper triangular } U \\ \text{pivots on the diagonal} \end{pmatrix}$$

Requirements: No row exchanges as Gaussian elimination reduces square A to U .

$$2. \quad A = LDU = \begin{pmatrix} \text{lower triangular } L \\ \text{1's on the diagonal} \end{pmatrix} \begin{pmatrix} \text{pivot matrix} \\ D \text{ is diagonal} \end{pmatrix} \begin{pmatrix} \text{upper triangular } U \\ \text{1's on the diagonal} \end{pmatrix}$$

Requirements: No row exchanges. The pivots in D are divided out to leave 1's on the diagonal of U . If A is symmetric then U is L^T and $A = LDL^T$.

$$3. \quad PA = LU \text{ (permutation matrix } P \text{ to avoid zeros in the pivot positions).}$$

Requirements: A is invertible. Then P, L, U are invertible. P does all of the row exchanges on A in advance, to allow normal LU . Alternative: $A = L_1 P_1 U_1$.

$$4. \quad EA = R \text{ (} m \text{ by } m \text{ invertible } E \text{) (any } m \text{ by } n \text{ matrix } A \text{) = rref}(A \text{).}$$

Requirements: None ! *The reduced row echelon form* R has r pivot rows and pivot columns, containing the identity matrix. The last $m - r$ rows of E are a basis for the left nullspace of A ; they multiply A to give $m - r$ zero rows in R . The first r columns of E^{-1} are a basis for the column space of A .

$$5. \quad S = C^T C = (\text{lower triangular}) (\text{upper triangular}) \text{ with } \sqrt{D} \text{ on both diagonals}$$

Requirements: S is symmetric and positive definite (all n pivots in D are positive). This *Cholesky factorization* $C = \text{chol}(S)$ has $C^T = L\sqrt{D}$, so $S = C^T C = LDL^T$.

$$6. \quad A = QR = (\text{orthonormal columns in } Q) (\text{upper triangular } R).$$

Requirements: A has independent columns. Those are *orthogonalized* in Q by the Gram-Schmidt or Householder process. If A is square then $Q^{-1} = Q^T$.

$$7. \quad A = X \Lambda X^{-1} = (\text{eigenvectors in } X) (\text{eigenvalues in } \Lambda) (\text{left eigenvectors in } X^{-1}).$$

Requirements: A must have n linearly independent eigenvectors.

$$8. \quad S = Q \Lambda Q^T = (\text{orthogonal matrix } Q) (\text{real eigenvalue matrix } \Lambda) (Q^T \text{ is } Q^{-1}).$$

Requirements: S is *real and symmetric*: $S^T = S$. This is the Spectral Theorem.

9. $A = BJB^{-1}$ = (generalized eigenvectors in B) (Jordan blocks in J) (B^{-1}).

Requirements: A is any square matrix. This *Jordan form* J has a block for each independent eigenvector of A . Every block has only one eigenvalue.

10. $A = U\Sigma V^T = \begin{pmatrix} \text{orthogonal} \\ U \text{ is } m \times m \end{pmatrix} \begin{pmatrix} m \times n \text{ singular value matrix} \\ \sigma_1, \dots, \sigma_r \text{ on its diagonal} \end{pmatrix} \begin{pmatrix} \text{orthogonal} \\ V \text{ is } n \times n \end{pmatrix}.$

Requirements: None. This *Singular Value Decomposition* (SVD) has the eigenvectors of AA^T in U and eigenvectors of $A^T A$ in V ; $\sigma_i = \sqrt{\lambda_i(A^T A)} = \sqrt{\lambda_i(AA^T)}$.

Those singular values are $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$. By column-row multiplication

$$A = U\Sigma V^T = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T.$$

If S is symmetric positive definite then $U = V = Q$ and $\Sigma = \Lambda$ and $S = Q\Lambda Q^T$.

11. $A^+ = V\Sigma^+ U^T = \begin{pmatrix} \text{orthogonal} \\ n \times n \end{pmatrix} \begin{pmatrix} n \times m \text{ pseudoinverse of } \Sigma \\ 1/\sigma_1, \dots, 1/\sigma_r \text{ on diagonal} \end{pmatrix} \begin{pmatrix} \text{orthogonal} \\ m \times m \end{pmatrix}.$

Requirements: None. The *pseudoinverse* A^+ has $A^+ A$ = projection onto row space of A and AA^+ = projection onto column space. $A^+ = A^{-1}$ if A is invertible. The shortest least-squares solution to $Ax = b$ is $x^+ = A^+ b$. This solves $A^T Ax^+ = A^T b$.

12. $A = QS$ = (orthogonal matrix Q) (symmetric positive definite matrix S).

Requirements: A is invertible. This *polar decomposition* has $S^2 = A^T A$. The factor S is semidefinite if A is singular. The reverse polar decomposition $A = KQ$ has $K^2 = AA^T$. Both have $Q = UV^T$ from the SVD.

13. $A = U\Lambda U^{-1}$ = (unitary U) (eigenvalue matrix Λ) (U^{-1} which is $U^H = \bar{U}^T$).

Requirements: A is *normal*: $A^H A = AA^H$. Its orthonormal (and possibly complex) eigenvectors are the columns of U . Complex λ 's unless $S = S^H$: Hermitian case.

14. $A = QTQ^{-1}$ = (unitary Q) (triangular T with λ 's on diagonal) ($Q^{-1} = Q^H$).

Requirements: *Schur triangularization* of any square A . There is a matrix Q with orthonormal columns that makes $Q^{-1}AQ$ triangular: Section 6.4.

15. $F_n = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_{n/2} \\ F_{n/2} \end{bmatrix} \begin{bmatrix} \text{even-odd} \\ \text{permutation} \end{bmatrix}$ = one step of the recursive FFT.

Requirements: F_n = Fourier matrix with entries w^{jk} where $w^n = 1$: $F_n \bar{F}_n = nI$. D has $1, w, \dots, w^{n/2-1}$ on its diagonal. For $n = 2^\ell$ the *Fast Fourier Transform* will compute $F_n x$ with only $\frac{1}{2}n\ell = \frac{1}{2}n \log_2 n$ multiplications from ℓ stages of D 's.